MGT 8040 DATA MINING

Dr. Frank Lee

OPTIMIZING PUBLICATIONS IN TOP JOURNALS

Final Project

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Executive Summary

Our project asks a simple question: Can we look at the features of a research paper and guess if it belongs in a high-impact journal or a low-impact one? This matters because sending a paper to the wrong journal can waste months of time and delay a researcher's career. We collected 104 papers from trusted databases and wrote down details like page count, word count, number of authors, paper type, and number of references.

We cleaned the data, fixed problems, and created new helpful features like "visual density" and "citation intensity." Then we turned impact factor into two groups: High and Low. We tested different models like GLM, logistic regression, decision tree, and random forest. Logistic regression worked the best because it correctly found most low-impact papers and avoided big mistakes.

The model showed that papers with more references, more pages, more words, and more figures were more likely to be high-impact. Papers in fields like AI also had a better chance. But the model was not good at finding high-impact papers, mostly because the dataset was small. In the future, collecting more papers and adding stronger features can help improve positive class recall and make the model more useful for real researchers.

1. Business Understanding

Academic publishing is a billion-dollar industry with lots of competition for placement in high impact journals. The ability to get published in one of these journals is strongly tied to an individual's career since it leads to promotions, larger grant allocations, research visibility, and many other outcomes. Misaligned submissions lead to long delays (3-6 month review cycles), wasted effort revising and resubmitting, and possible increased administrative burden. With all of this in mind, choosing the proper journal is extremely important to any researcher. This led us to the research question at hand:

"Can measurable features of a scholarly paper reliably classify it as belonging to high impact or low impact journals?"

Stakeholder	Why They Care	Pain Points	How Our Model Helps
Graduate Students / PHD Candidates	Need for publications for career progression	Rejections delay graduation; unclear journal targeting	Provides data driven journal selection guidance and paper improvement insights
Faculty Researchers	Publication success influences tenure, promotion, grants	Wasted time due to mismatched journal submissions	Helps prioritize journals and optimize paper structure
Universities / Research Institutions	Value high impact publications for rankings and prestige	Uneven publication performance across departments	Supports training programs and enhances research efficiency

Figure 1: Stakeholder Analysis

Impact factor is widely used as a main criteria for assessing research quality, so it serves as the best target variable for this classification task.

The academic publishing ecosystem involves multiple stakeholders whose decisions and outcomes are influenced by journal impact and publication success. Figure 1 summarizes key stakeholders and their interests.

Our success criteria reflect the real world costs of journal targeting decisions. Since incorrectly predicting a paper as suitable for a high impact journal leads to lost time and delayed publications, our evaluation emphasizes precision and overall model reliability. Table 2 presents the specific metrics and target values guiding our assessment.

Metric	Target	Reasoning
Specificity (1-FPR)	≥ 0.75	High cost of false positives; main success criteria
Accuracy (Total)	≥ 0.70	Ensures model still captures true positives and true negatives
AUC	≥ 0.70	Shows strong separation ability

Figure 2: Success Criteria

We hope that researchers and institutions will be able to utilize our model to avoid waste and improve publication outcomes. For researchers, this will be done by tailoring their writing to improve their competitiveness. This doesn't necessarily mean that individuals should completely change their research to achieve high impact publication, but instead guide them to appropriate journal submissions. As for institutions, our findings can support mentoring programs and help departments better guide early career researchers through the publication process. This can ultimately strengthen overall research visibility and improve institutional standing.

2. Data Understanding

In order to collect our data we visited many different academic journal databases. Of those various databases, the primary one used was ScienceDirect. Other databases include: mdpi, emerald, ieee, jmir, and tandfonline. The papers selected to be used in our analysis fell within the following domains: Business, Healthcare, and Social Sciences, with some papers having a combination of one of these domains alongside AI. For each paper, we manually extracted variables using a standardized extraction method to ensure consistency. The count of table and figures were combined, page ranges recorded uniformly, word count collected through Microsoft Word "word count" tool, and paper types clearly agreed upon to ensure they remain consistent across group members.

ScienceDirect and other databases are highly reputable, however there are some potential biases that need to be addressed. There is possible topic bias since different fields inherently have different impact factor norms. Additionally, there is a paper type imbalance within the dataset, with most paper types belonging to experimental and review. We attempted to diversify the paper types, however it was difficult since commentary pieces were difficult to come by.

2.1 Data Description

Variable	Meaning / Description	Data Type	Relevance to Business Problem	Data Quality
Impact Factor	Numeric rating of journal prestige	Continuous	Target variable for classification into High/Low Impact	Variation across fields; 0% missing
Page Count	Total number of pages in the paper	Integer	Longer papers may reflect depth associated with high impact venues	0% missing
Word Count	Estimated total word count	Integer	Strong measure of paper length/detail	0% missing
Number of Authors	Number of authors listed on the paper	Integer	Collaboration patterns can influence journal targeting	0% missing
Number of figures	Count of table and figures	Integer	Reflects methodological quality	0% missing
Paper Type	4 types: review, experimental, conceptual, commentary	Categorical	Different paper types align with different journal tiers	0% missing
Number of references	Total number of cited references in the paper	Integer	Higher reference count may reflect depth	0% missing

Figure 3: Summary of Variables

The variables listed in Figure 3 represent the features extracted from academic papers across the multiple databases listed previously. Each was chosen for its potential contributions to distinguishing between high impact and low impact journals

Several variables in our dataset exhibit natural outliers and skewed distributions. For example, impact factor values are right skewed due to a small number of exceptionally high impact journals. Page count, word count, and number of references also show right skewed patterns, with review papers in particular affecting word counts. The number of authors varies widely across domains, producing occasional large author groups. These distributions are expected and will be accounted for during modeling and evaluation.

2.2 Exploratory Data Analysis

Before modeling, we performed an exploratory data analysis to examine how the paper features vary and how they may relate to the journal impact level. The goal of this EDA is to identify patterns, assess data quality, and determine which variables may be most informative for classification.

First we want to look at basic descriptive statistics of key numerical variables within the dataset (See Figure 4). The variables are showing signs of having possible outliers that skew their distributions. Impact factor has a mean of 5.5, however there is an entry with an extremely large impact factor sitting at 21.6. This trend is applicable to all the variables depicted in the table, since their maximum values all lie way beyond their respective means. The number of figures with a mean of 8.6 and

standard deviation of 6.1 suggests that there is large variability in how visual papers are. For page count, that range of 6 to 50 may differentiate short commentaries versus long review articles, suggesting that page count may correlate with paper type. Author count is showing a mean of 3.75 authors, which is expected for most research papers. We expect that the maximum value of 17, and other larger counts of authors, is attributed to our healthcare domain papers. For reference count, a mean of 78 suggests that most papers are well cited, however a standard deviation of 49 seems very large. This possibly suggests that reference counts vary drastically by paper type (i.e. review has a large amount of references versus conceptual/commentary).

Variable	Mean	SD	Min	Max
Impact Factor	5.538	3.637	0	21.6
Number of Figures	8.567	6.148	0	29
Page Count	18.721	9.871	6	50
Word Count	9473.279	5811.661	1300	49000
Author Count	3.750	2.783	1	17
Number of References	77.798	48.719	0	270

Figure 4: Descriptive Statistics

With the numerical patterns established, we next examine the categorical variables and class distribution to better understand how the papers are grouped within the dataset. First, we want to examine the class distribution (high versus low impact), which was split at an impact factor of 7.

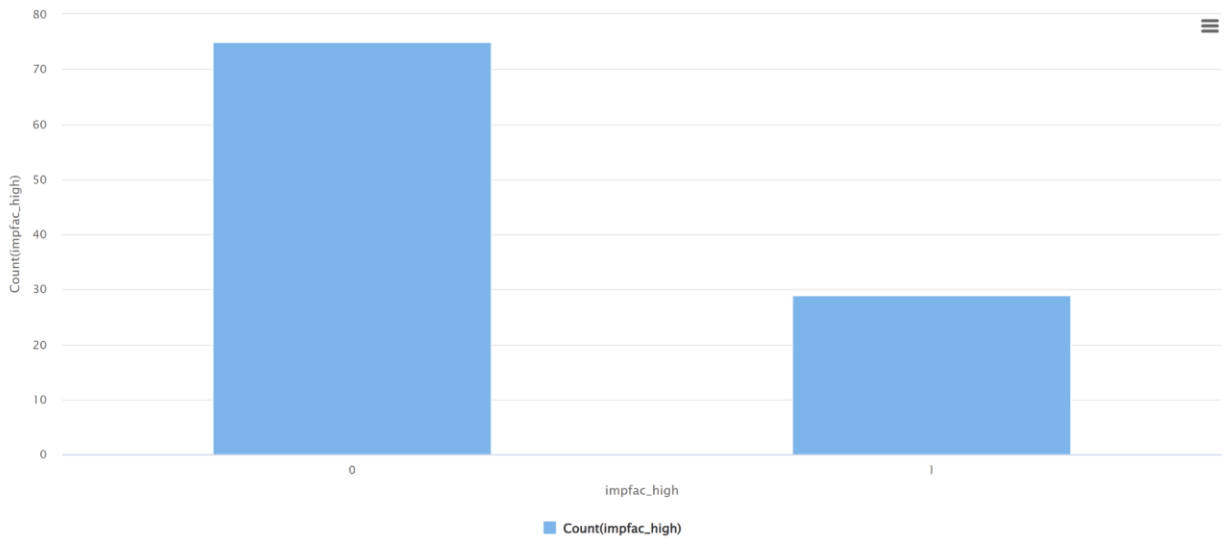


Figure 5: Barplot of Target Class Frequency

In figure 5, most values fall into the low impact class (0 = low impact, 1 = high impact), with low impact containing 75 entries and the remaining 29 with high impact labels. This moderate imbalance suggests that the model may need to be evaluated carefully to make sure performance is not driven by the larger class.

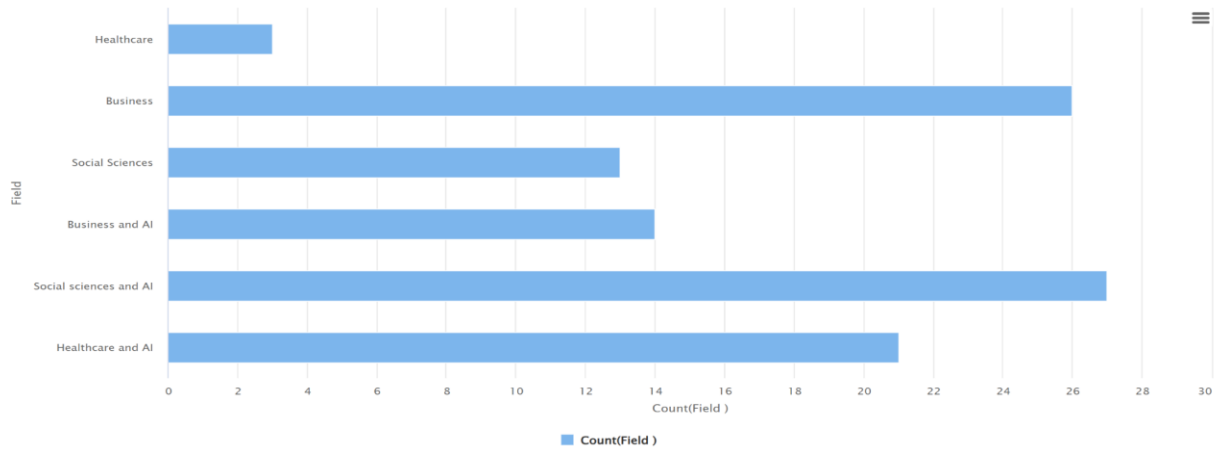


Figure 6: Barplot of Paper Domains Frequency

The field distribution shows that most papers fall into mixed domains, particularly Social Sciences and AI (27 papers) and Healthcare and AI (21 papers). Business (26 papers) and Business and AI (14 papers) also make up a large portion of the dataset, while pure Healthcare is less represented with only 3 papers. This uneven distribution suggests that some fields contribute more to the dataset than others, which may influence patterns observed later in the analysis.

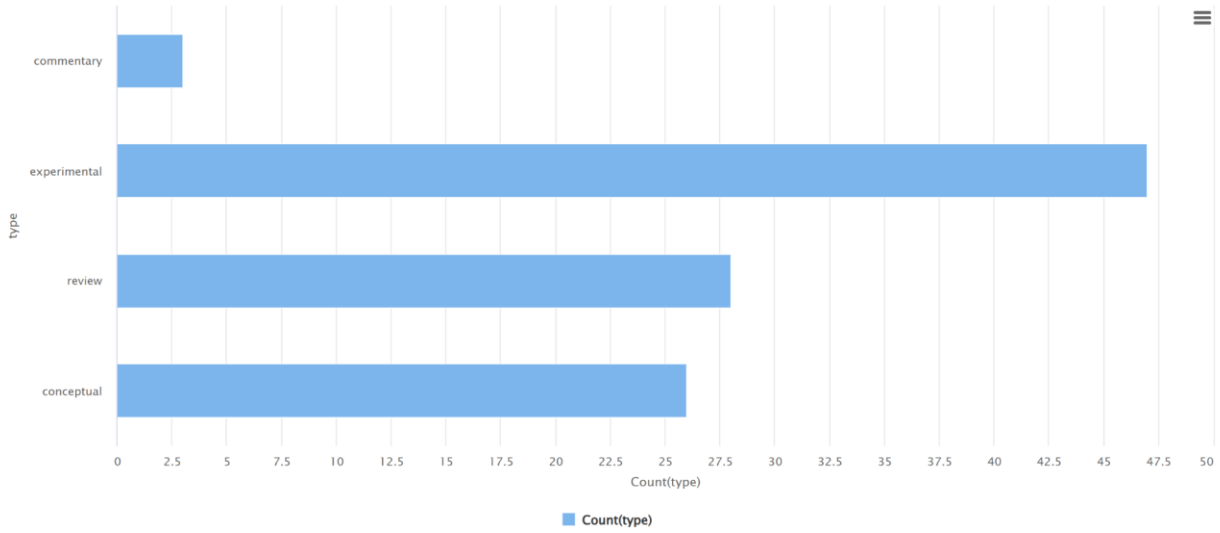


Figure 7: Barplot of Paper Types Frequency

Experimental papers make up the largest share of the dataset with 47 entries, while reviews (28 papers), and conceptual papers (26 papers) appear in similar amounts. Commentary papers are rare, with only 3 entries, showing a clear imbalance in paper types.

The correlation matrix (Fig 8) shows strong relationships among the length related variables. Pages, words, figures, and references are all moderately to strongly correlated, which is expected since longer papers typically include more content. On the other hand, impact factor shows only very weak correlations with all numerical features, with values close to zero. This suggests that the simple linear relationships are not sufficient to distinguish high impact from low impact journals and reinforces the need for predictive classification modeling.

Attribu...	figures	pages	words	authors	referen...	impfac...
figures	1	0.400	0.540	0.036	0.211	0.007
pages	0.400	1	0.629	-0.096	0.387	0.029
words	0.540	0.629	1	-0.023	0.547	0.077
authors	0.036	-0.096	-0.023	1	0.186	0.121
references	0.211	0.387	0.547	0.186	1	0.165
impfactor	0.007	0.029	0.077	0.121	0.165	1

Figure 8: Correlation Matrix

The boxplot comparing reference counts across high and low impact papers shows that high impact papers tend to include more references on average. Although that upper tail is larger for low impact papers, we can see that the high impact papers always have references. This suggests that reference volume may contain some predictive signal for classification.

The boxplots (Fig 9) for word count, page count, and number of figures all show a similar pattern. In each case, the high impact group has a higher average value than the low impact group. Interestingly, the paper with an extremely large word count falls into the low impact class. This suggests that yes, depth of analysis is important, but also the quality of analysis is key. Although there is overlap between the two classes, the general trend suggests that papers published in high impact journals tend to be longer and include more visual content.

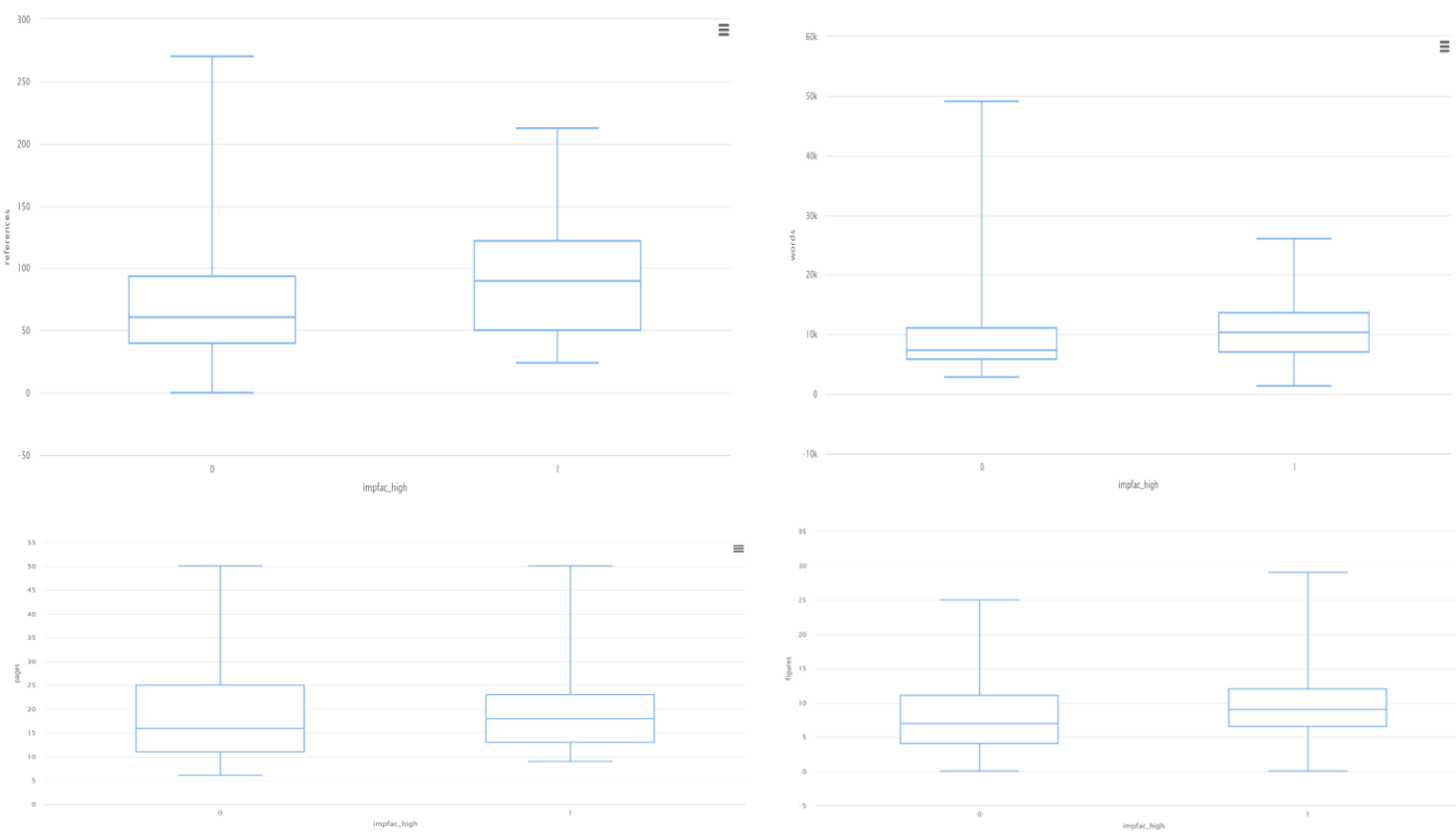


Figure 9: Boxplots of References (Top L), Words (Top R), Pages (Bottom L), Figures (Bottom R) w.r.t Impact

The EDA reveals several patterns that could contribute to journal impact prediction, such as higher averages in word count, page count, references, and figures for high impact papers. However, the weak correlations with impact factor indicate that these relationships are not straightforward. In the following section, we describe the modeling framework we will use to evaluate whether paper features can reliably predict journal impact level.

3. Data Preparation

3.1 Data Cleaning

In this step, we made sure the dataset was neat and ready to use. First, we found one paper that had 0 references, and this caused a problem when we took the log of the references, because log(0) becomes infinity. Since this value could not be used, we removed that row, leaving us with 103 papers.

We also checked that all variables had the correct data types. For example, the impact factor was stored as a number at first, but we later changed it into two groups; High and Low. This helped turn the problem into a classification task. We also checked that new variables and interaction terms did not create strange or inconsistent numbers. For outliers in references, we applied log to it.

3.2 Feature Engineering

Here, we created new variables that might help the model understand the papers better. For example, we made a variable called visual density by dividing the number of figures by the number of pages. This shows how "visually heavy" a paper is. We also made a new feature called citation intensity, which tells us how many references a paper has for every 1000 words.

We turned the field variable into dummy variables in Turbo Prep. Even though Auto Model normally makes dummies automatically, we needed to do it ourselves so we could build an interaction term between the field and the number of words.

We also changed the impact factor into two bins:

- High (above 7)
- Low (7 or below)

This cutoff came from what many people in the business journal world consider "high impact."

Attribut...	authors	bus_x_...	busai_x...	citation...	Field = ...	Field = ...	Field = ...	Field = ...	Field = ...	figures	healthai...	imp_fac...	page
ref_log	0.127	0.089	0.271	0.379	-0.129	0.192	-0.032	0.072	0.001	0.196	0.024	-0.204	0.384
ss_x_wo...	-0.258	-0.178	-0.128	-0.078	-0.209	-0.146	-0.186	0.969	-0.220	0.057	-0.167	0.007	0.569
ssai_x_...	-0.017	-0.221	-0.159	-0.136	-0.259	-0.181	-0.231	-0.174	0.766	0.221	-0.207	-0.165	0.273
type = co...	-0.134	-0.153	-0.016	-0.053	-0.162	-0.026	0.051	0.194	-0.028	-0.158	0.024	0.153	0.067
type = ex...	0.003	0.232	-0.080	-0.208	0.300	-0.022	-0.270	0.004	-0.014	0.278	-0.196	-0.120	0.006
type = re...	0.191	-0.135	0.128	0.324	-0.244	0.076	0.287	-0.167	0.033	-0.084	0.226	-0.054	-0.02
visual_d...	0.144	0.194	0.073	-0.093	0.170	0.130	-0.130	-0.224	0.031	0.728	-0.113	-0.079	-0.21

Figure 10: Correlation Matrix After Data Transformation

While selecting features, we removed variables that were too similar to each other. For example, the figure variable was removed because it had more than 0.7 correlation with visual density. The interaction terms for social science fields were also removed because they were too strongly related to the field variable.

3.3 Data Transformation

We used a log transformation on the references variable (ref_log) because it had a long tail with very big numbers, and the log made the histogram more balanced and less skewed.

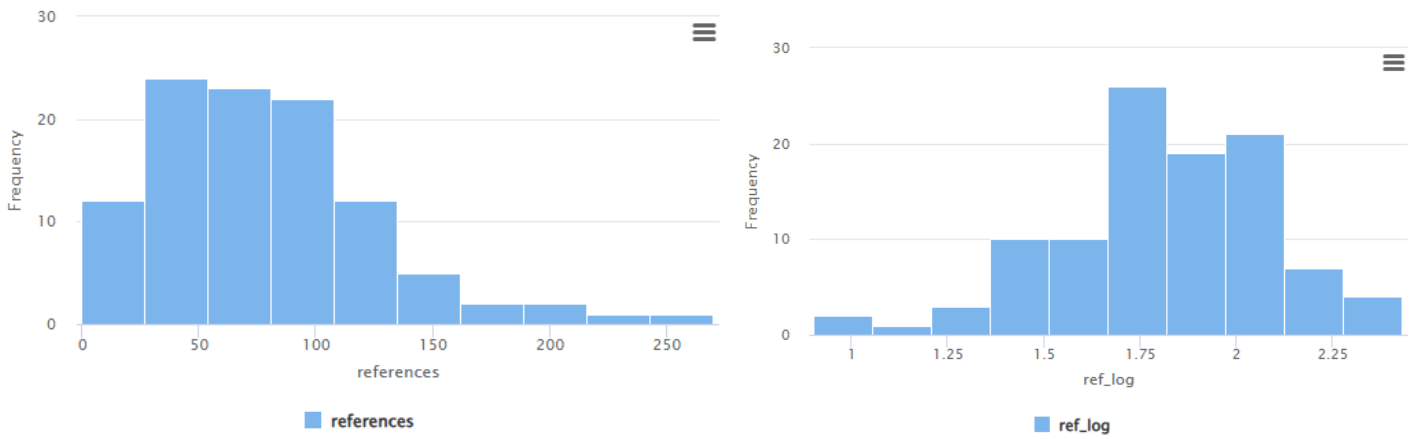


Figure 11: Skewed References to Unskewed Log_References

We also handled the categorical variables by turning the field of the paper into dummy variables and turning the impact factor into High/Low groups. These transformations helped the model understand categories in a simple way.

We did not use any fancy dimensionality reduction methods, but we did reduce the number of variables by removing those that were too similar. This made the dataset cleaner and easier for the model to learn from.

4. Model Selection

4.1 Models

As a group of paper writers, we wanted to use models that help us understand what kinds of papers usually end up in high-impact journals. So we picked models that are not only strong but also easy to explain.

1. GLM (Generalized Linear Model)

We used GLM because it is simple and lets us see which features of a paper help increase impact. GLM helps show clear relationships, like whether more citations or more pages make a paper more likely to be “High Impact.”

2. Logistic Regression

We chose logistic regression because it is very interpretable. It tells us how each variable changes the chances of a paper becoming high impact. As paper writers, we care about understanding *why* something matters, so logistic regression helps us learn which writing choices matter most.

3. Decision Tree

We included a decision tree because it makes rules we can actually understand, almost like a checklist. It is also widely recognised method for classification problems. It can show simple paths such as "if citation intensity is high, more likely High Impact." This helps us see practical steps we can take as writers.

4. Random Forest

Random forest is powerful because it builds many trees and combines them, so it finds patterns that simple models miss.

4.2 Cost-Benefit Analysis

Since we are looking at this problem from the viewpoint of paper writers, we created a cost matrix that reflects how good or bad each outcome is for *us*, not for a journal.

Here is the cost matrix:

Cost Matrix	True High	True Low
Predicted High	10000	-10000
Predicted Low	-7000	1000

Figure 12: Cost Benefit Matrix

Why these dollar values make sense for us as paper writers:

- +10000 (True High predicted as High):
  If our paper truly belongs in a high-impact journal and the model predicts that, this is a huge win. It means our effort, time, and reputation pay off. We treat this like a big reward.
- –10000 (False Positive):
  This is the worst-case scenario for us. It means we think our paper is strong enough for a high-impact journal, but we misjudge it. We might waste months sending it to top journals and getting rejected. That feels like a huge "loss."
- –7000 (False Negative):
  This scenario also hurts because we might underestimate ourselves and send the paper to a lower-tier journal even though it had high-impact potential. We lose recognition and visibility, but it's not as damaging as wasting months on top-tier rejections.
- +1000 (True Low predicted as Low):
  If the model correctly says our paper isn't ready for a high-impact journal, we save time by improving it first or submitting to a more realistic place. That's a small but meaningful win.

5. Evaluation

5.1 Performance Matrix

As to evaluate from a researcher's perspective for papers publication success for areas such as AI, Social Sciences and Business for **103 research papers**, we used multiple models for comparison. Models were compared on the basis of accuracy, specificity, AUC, standard deviation, gains and overall performance.

Model	Accuracy	Specificity	AUC	SD	Gains	Total Time	Train Time	Test Time
Generalized Linear Model	0.47	0.39	0.68	0.17	-41000	457	282	190
Logistic Regression	0.70	0.96	0.65	0.09	-11000	377	223	143
Decision Tree	0.60	0.86	0.50	0.13	-33000	457	136	71
Random Forest	0.60	0.71	0.61	0.08	-15000	1248	165	167

Figure 13: Performance Matrix

Logistic regression demonstrates the best accuracy = 0.70 i.e. 7 out of 10 cases were correctly classified with a specificity = 0.96 i.e. model's ability to identify papers correctly which are likely to achieve success or not based on metrics. Though Generalized linear model has highest AUC=0.68, the best model would be logistic regression based on gains and overall performance. Given the models performance, Logistic Regression is best and relaible model.

5.2 Confusion Matrix Analysis

	true Low	true High	class precision
pred. Low	21	8	72.41%
pred. High	1	0	0.00%
class recall	95.45%	0.00%	

Figure 14: Confusion Matrix

The confusion matrix predicted 21 true low predicted low, 8 false negatives, 1 false postive and 0 true high predicted high. Following are the observations form confusion matrix*:*

- **True low, predicted low:** The model accurately predicted the papers which has low impact potential. It demonstartes the importance of features and how correct identification of low-impact paper can help improve the content of paper before publication and save time.
- **False negatives:** The model misclassfied certain high impact papers as low impact papers. This shows potential loss for authors as misclassification can lead to loss of time, money and resources. Writers may submit paper with potential to get published in high impact journal in low impact journal. This aligns with cost benefit analysis showing cost of false negative as -$7,000.
- **False positive:** There was only 1 case of low impact paper predicted as high impact paper. This shows that model is conservative and minimizes the waste of time, money and resources.

- **True high, predicted high:** Our model did not identify any paper as high impact paper which is incorrect and explains zero sensitivity. It could be because of small dataset, skewed distribution or thresholding used during feature engineering.

Author needs to be cautions while relying solely on predictions for paper submission. However this model will help authors to prioritize their resources, time management and help focus on improvements. Model is effective for filtering out low-impact papers but is less reliable for high impact papers.

5.3 Lift Chart Analysis

Analysis on model probabilities shows strong specificity which means that high proportion of correctly classified low-impact papers forms the top deciles. Lower deciles which predicted high-impact papers has mostly misclassified low-impact papers that indicates that the model is not effective at identifying high-impact papers. However, the lift curve exhibit the utility to screening out weak papers, but the limitation is it's ability to identify highly successful ones; lift curve shows limited separation between both.

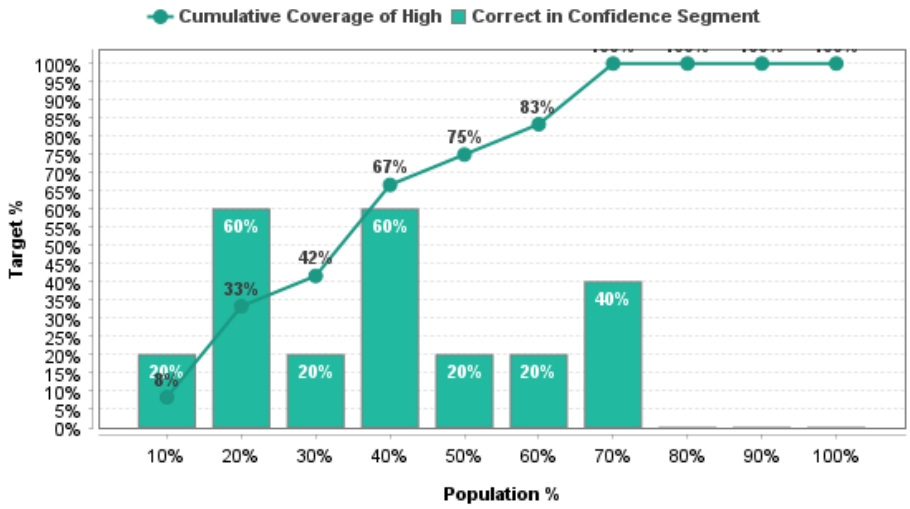


Figure 15: Lift Chart

5.4 ROC Curve Analysis

From the classification model's ROC curve for logistic regression model AUC=0.65 which indicating that it performs poorly. It also suggests that this model can distinguish between high-impact papers and low-impact papers 65% of time; better than random forest (0.61) and decision tree (0.50) still shows the limited predictive ability.

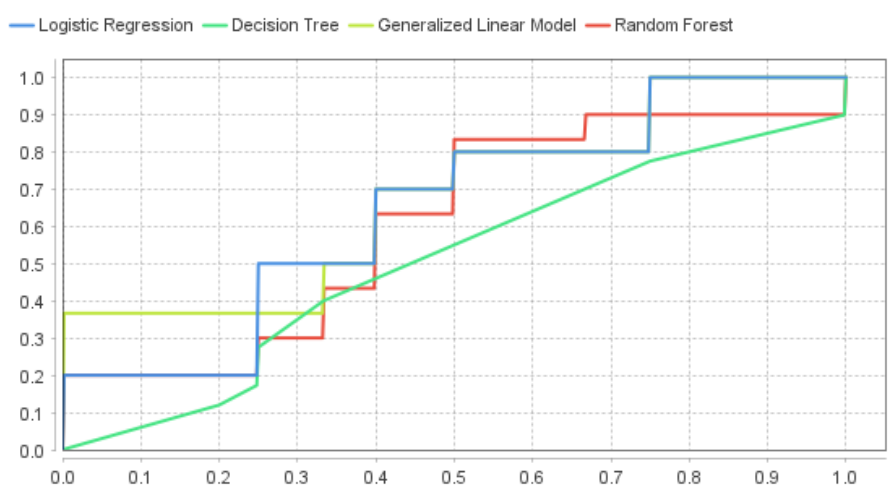


Figure 16: ROC Curves

However, value is in line with the overall behavior of model in the confusion matrix, where it has high specificity of 0.96. The model is highly reliable to screen low-impact papers but it is unable to identify

high-impact papers effectively. The AUC illustrates the model as conservative because it correctly classifies papers with low impact however, fails to identify high-impact papers.

6. Deployment and Conclusion

6.1 False Positive/Negative Analysis

For successful deployment, it is important to understand how the model's misclassifications can affect author's decision-making. Incorrect predictions significantly have different repercussions than that what a journal would prioritize as the model was developed from the writer's point of view. For instance, even one **false positive** i.e. low-impact as compared to expected high-Impact paper can be a costly error (–$10,000). While, a false positive makes authors to believe that their paper needs improvement to be published in a high-impact journal. For paper submission to high-impact journal which usually have long review cycles could have adverse effect on emotional well-being and can create reputational strain in case of repeated rejections.

Strategies for Mitigation: Model's output along with qualitative review can help in determining features for publication in high impact journal.

False negative: There were 8 cases where high impact was predicted as low impact. We can reduce the chances of optimistic misclassification through ensemble models. This mistake can cost a lot (–$7,000 per case) and make authors undervalue strong paper. This can possibly lead to submission of potentially high impact papers to low-impact journals making it less visible. **Strategies for Mitigation:** We can strengthen features which captures attention of scholars and institutions. We can address the class imbalance through oversampling technique and by lowering high-impact classification threshold increase sensitivity.

To summarize errors, model is conservative and therefore we should focus on screening and not on selection for deployment.

6.2 Model Interpretability and Insights

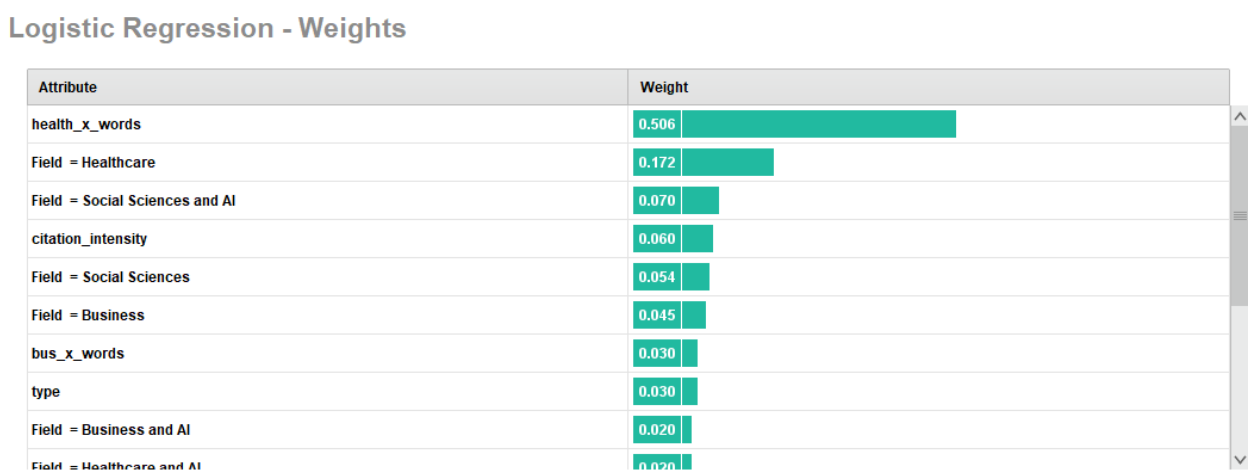


Figure 17: Logistic Regression Insights

Transparency and performance evaluation were made possible by chosen models: Generalized Linear Model, Logistic Regression, Decision Tree and Random Forest. Interpretability analyses revealed a number of useful insights as following:

- Citation Intensity which is references per 1000 words strongly correlates with impact level demonstrating papers that have a lot of references usually have a stronger theoretical foundation and are perceived as more sophisticated.
- Visual Density-figures /pages also contributes to success, particularly in fields like AI, business, social sciences and healthcare where empirical representation and visual clarity are important.
- The positive correlation between high impact paper and length (pages and word count) suggests that more comprehensive papers are valued.
- Field differences are important, among all papers with AI content demonstrates higher likelihood of becoming high-impact.
- Field and Word Count and other interactive terms show how writing strategies that work in one field may not work the same in another.

To address these issues, authors should increase citation, especially in AI and Healthcare. Write structurally rich papers which has more figures in consonance to pages. Authors in business concentration may require clarity and focus as compared to length while in AI concentration researchers should create longer and more in-depth papers. Instead of using model predictions as a tool for making a decision, we should use them to check for quality.

6.3 Future Work and Limitations

Limitations: The model cannot yet identify papers with a high impact if its sensitivity is zero. Also, dataset which is 103 papers, is small and imbalanced and covers only few high-impact cases. Information on some of the impactful features like quality of abstract, author's reputation, institutional ranking or cite score were not available for all the papers. Since target class is uncommon, it contributed to the 0 true-positive.

For **future work** we can collect more papers from a variety of journals and years to expand the dataset. Include scholarly influence features such as h-index, citations per paper, funding data or metrics of collaboration network. We will use advance models to increase sensitivity such as Gradient boosted trees. To enhance high-impact detection, we can use penalized logistic regression to rebalance classes. For higher accuracy we need to combine model predictions with expert or peer review.

7. Appendix

S. No. Number	paper_url Category	year Number	title Category	Field Category	figures Number	pages Number	citations Number	words Number	autho Number
1	https://www.sci...	2024	Consolidated ...	Healthcare and...	3	12	8	6069	17
2	https://www.sci...	2024	The influence o...	Healthcare and...	0	7	1.100	3156	2
3	https://www.sci...	2024	Building reliabl...	Social Science...	14	12	20.700	5686	2
4	https://www.sci...	2024	Regulating gen...	Social Science...	7	24	18.400	5687	2
5	https://www.sci...	2024	An adaptive net...	Healthcare and...	24	30	6.100	15689	5
6	https://www.sci...	2024	How to identify ...	Healthcare and...	9	14	10.200	7060	4
7	https://www.sci...	2024	Evaluating the ...	Social Science...	12	12	0	6328	9
8	https://www.sci...	2024	Me, myself and ...	Social Science...	7	20	26.300	10338	4
9	https://www.sci...	2024	Understanding ...	Social Science...	8	45	10.900	13730	7

104 rows - 16 columns (5 nominal, 11 numerical)

COMMIT GENERATE CLEAR ALL CANCEL

Name: citation_intensity

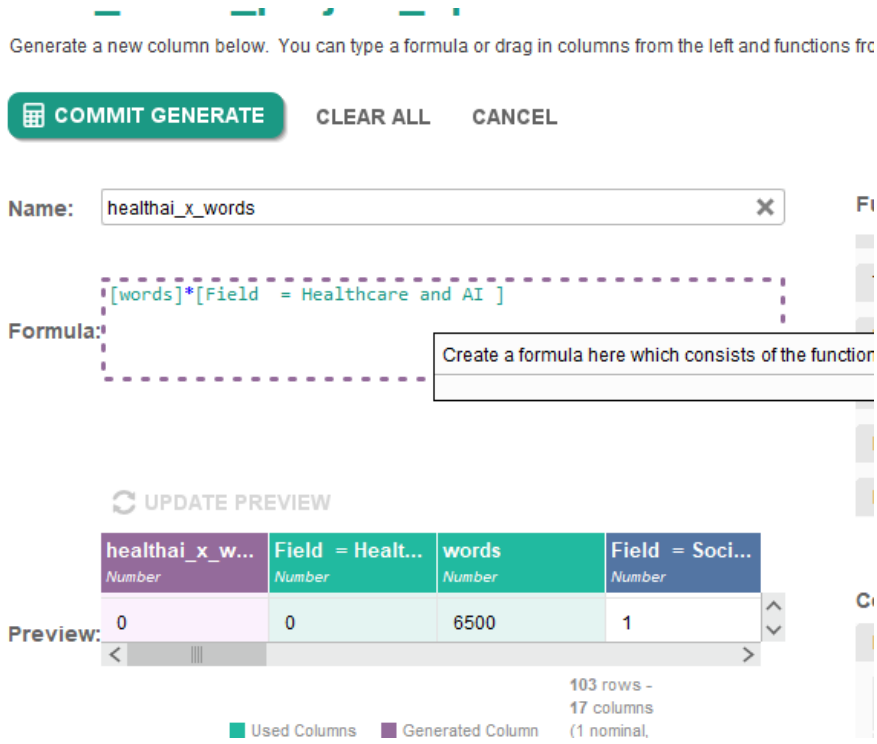
Formula:

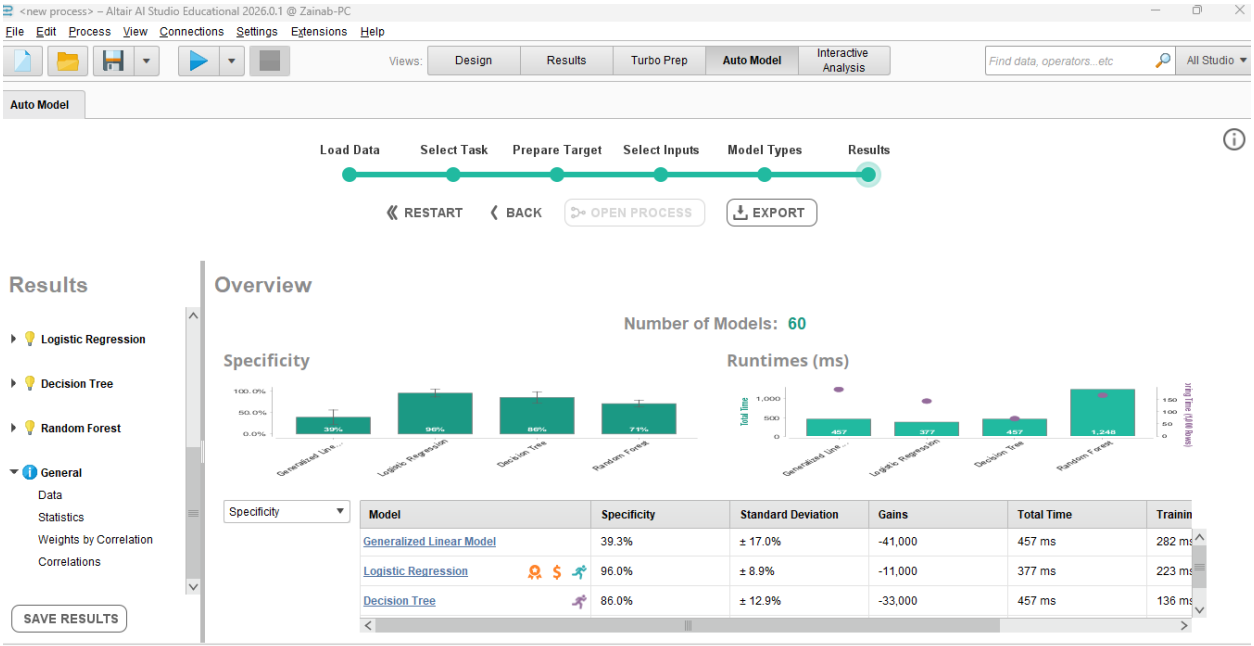
```
[references]/([words]/1000)
```

UPDATE PREVIEW

Preview:

citation_inte... Number	references Number	words Number	ref_log Number
6.462	42	6500	1.623


Generate a new column below. You can type a formula or drag in columns from the left and functions fro
COMMIT GENERATE
CLEAR ALL
CANCEL
Name:
healthai_x_words
Formula:
[words]*[Field = Healthcare and AI]
Create a formula here which consists of the function
UPDATE PREVIEW
Preview:
healthai_x_w... Number
Field = Healt... Number
words Number
Field = Soci... Number
0
0
6500
1
103 rows -
17 columns
(1 nominal,
Used Columns
Generated Column


<new process> – Altair AI Studio Educational 2026.0.1 @ Zainab-PC
File Edit Process View Connections Settings Extensions Help
Views: Design Results Turbo Prep Auto Model Interactive Analysis
Find data, operators...etc
All Studio
Auto Model
Load Data
Select Task
Prepare Target
Select Inputs
Model Types
Results
RESTART
BACK
OPEN PROCESS
EXPORT
Results
Logistic Regression
Decision Tree
Random Forest
General
Data
Statistics
Weights by Correlation
Correlations
SAVE RESULTS
Overview
Number of Models: 60
Specificity
Runtimes (ms)
Specificity
Model
Specificity
Standard Deviation
Gains
Total Time
Generalized Linear Model
39.3%
± 17.0%
-41,000
457 ms
Logistic Regression
96.0%
± 8.9%
-11,000
377 ms
Decision Tree
86.0%
± 12.9%
-33,000
457 ms

Logistic Regression - Model

Attribute	Coefficient	Std. Coefficient	Std. Error	z-Value	p-Value ↑
bus_x_words	0.000	2.336	0.000	2.059	0.040
visual_density	-4.083	-1.369	2.295	-1.779	0.075
authors	0.427	1.157	0.266	1.606	0.108
pages	-0.113	-1.123	0.079	-1.424	0.154
Field = Social Sciences a...	11.467	4.917	9.983	1.149	0.251
Field = Social Sciences	9.624	3.424	10.209	0.943	0.346
citation_intensity	0.107	1.220	0.120	0.895	0.371
Field = Business and AI	8.037	2.703	10.299	0.780	0.435
healthai_x_words	0.001	2.008	0.001	0.769	0.442

Logistic Regression - Model

Attribute	Coefficient	Std. Coefficient	Std. Error	z-Value	p-Value ↑
healthai_x_words	0.001	2.008	0.001	0.769	0.442
Field = Business	5.429	2.328	10.054	0.540	0.589
words	0.000	0.323	0.000	0.353	0.724
busai_x_words	0.000	0.371	0.000	0.277	0.782
ref_log	0.131	0.035	3.760	0.035	0.972
Intercept	-19.105	-10.990	854.196	-0.022	0.982
Field = Healthcare	59.637	11.266	2768.936	0.022	0.983
health_x_words	-0.009	-9.718	0.464	-0.019	0.985
type.experimental	10.081	10.081	854.134	0.012	0.991
type.conceptual	9.094	9.094	854.136	0.011	0.992

Logistic Regression - Weights

Attribute	Weight
health_x_words	0.506
Field = Healthcare	0.172
Field = Social Sciences and AI	0.070
citation_intensity	0.060
Field = Social Sciences	0.054
Field = Business	0.045
bus_x_words	0.030
type	0.030
Field = Business and AI	0.020
Field = Healthcare and AI	0.020

type	0.030
Field = Business and AI	0.020
Field = Healthcare and AI	0.020
pages	0.017
busai_x_words	0.013
ref_log	0.012
healthai_x_words	0.008
authors	0.007
visual_density	0.007
words	0.006